E-Commerce Customer Purchase Analysis

Project Documentation

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1. Problem Statement

An online shopping platform records information about customer purchases including customer details, product purchased, quantity, price, and city location. The marketing department wants to analyze purchasing behavior to improve promotional campaigns and identify valuable customers.

The key business questions that need to be answered are:

- Which customers spend the most money?
- Which products are most popular among buyers?
- Which cities generate the most revenue for the platform?

The challenge is to analyze a dataset of 827 customer purchase records and generate meaningful insights using SQL, Python, and Excel tools to support data-driven marketing decisions.

2. Solution & Features

The project provides a complete end-to-end data analysis pipeline using three different tools — SQL, Python, and WPS Excel — to answer all the business questions. The solution is divided into four main components:

2.1 Dataset

A CSV dataset of 827 customer records was generated containing the fields: customer_id, customer_name, city, product, quantity, and price. The dataset covers 10 Indian cities (Pune, Mumbai, Delhi, Nagpur, Bangalore, Hyderabad, Chennai, Kolkata, Jaipur, Ahmedabad) and 10 product categories (Mobile, Laptop, Tablet, Headphones, Smartwatch, Camera, Speaker, Keyboard, Monitor, Printer).

2.2 SQL Analysis

SQL queries were written and executed using Python's built-in SQLite library to answer all four analytical tasks:

- Display all purchase records with computed total_purchase column
- Calculate total spending per customer using GROUP BY and SUM
- Identify the most purchased product by total units sold
- Calculate city-wise revenue with total order counts

2.3 Python Analysis

A Python script using the pandas library was developed to perform the following tasks programmatically:

- Load the dataset from CSV using pandas
- Create the total_purchase column (quantity x price)
- Identify top 10 highest-spending customers
- Analyze product popularity by total units sold
- Generate bar chart and pie chart visualizations using matplotlib

3. Screenshots & Visualizations

The following visualizations were generated using the Python matplotlib library from the 827-record dataset. The charts clearly answer the key business questions about product sales and city-wise revenue distribution.

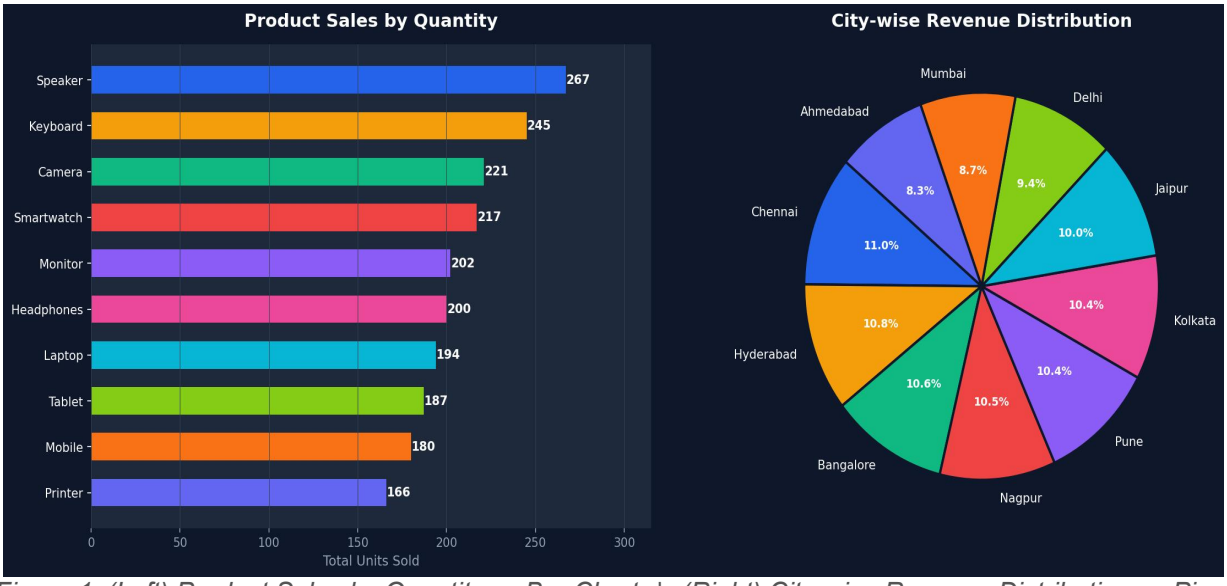

Figure 1: (Left) Product Sales by Quantity — Bar Chart | (Right) City-wise Revenue Distribution — Pie Chart

Key Observations from Visualizations

- **Top Products:** Speaker (267 units) and Keyboard (245 units) are the most purchased products by quantity.
- **Revenue Leaders:** Chennai (40.2L) and Hyderabad (39.5L) generate the highest revenue among all cities.
- **Balanced Distribution:** The pie chart shows a near-equal revenue distribution across all 10 cities, indicating no single city dominates.
- **High-Value Products:** Despite lower quantity, Laptop contributes significantly to revenue due to its high unit price of 60,000.

WPS Excel — Pivot Table Summary

A pivot table was created in WPS Excel showing product-wise revenue broken down by city. Additional charts (bar and pie) were inserted directly into the Excel workbook using the built-in charting features.

SQL Query Results Summary

City	Total Revenue (INR)	City	Total Revenue (INR)
Chennai	40,22,000	Kolkata	38,11,500
Hyderabad	39,47,500	Jaipur	36,71,500
Bangalore	38,97,000	Delhi	34,39,500
Nagpur	38,71,500	Mumbai	31,87,000
Pune	38,26,500	Ahmedabad	30,42,000

4. Tech Stack

Tool / Technology	Version / Details	Purpose
Python	3.x	Core programming language
pandas	Latest	Data loading, manipulation, analysis
matplotlib	Latest	Data visualization (charts)
SQLite3	Built-in	In-memory SQL query execution
openpyxl	Latest	Excel file creation and formatting
WPS Excel	Latest	Pivot tables and charts
VS Code	Latest	Code editor and terminal
CSV	827 records	Raw dataset storage format

5. Unique Points

- **Multi-tool Approach:** The project solves the same analytical problem using three different tools (SQL, Python, Excel), demonstrating versatility and reinforcing concepts across different platforms.
- **Large Realistic Dataset:** A dataset of 827 customers was generated with realistic Indian names, cities, and product categories to simulate a real-world e-commerce scenario.
- **Dynamic Excel Formulas:** The Excel file uses live formulas (=E2*F2) rather than hardcoded values, making it fully dynamic and recalculate automatically when data changes.
- **In-Memory SQL:** SQL queries are executed using Python's built-in SQLite3 library with no external database required, making the project self-contained and portable.
- **Automated Pivot Table:** A complete pivot table of Product x City revenue was generated programmatically with row and column totals using openpyxl, replicating WPS Excel pivot functionality in code.
- **Professional Visualizations:** Charts use a dark theme with a carefully selected color palette for clear, visually appealing output suitable for business presentations.

6. Future Improvements

- **Interactive Dashboard:** Build a web-based dashboard using Streamlit or Power BI to allow the marketing team to filter and explore data interactively.

- **Customer Segmentation:** Apply clustering algorithms (K-Means) to segment customers into groups such as high-value, occasional, and low-spend buyers for targeted marketing campaigns.
- **Time Series Analysis:** Add a purchase_date column to the dataset and analyze seasonal trends, monthly revenue patterns, and peak buying periods.
- **Product Recommendation:** Implement a basic recommendation system to suggest products to customers based on their purchase history and city.
- **Real Database Integration:** Replace the in-memory SQLite database with a real MySQL or PostgreSQL database to handle larger datasets and support concurrent queries.
- **Automated Reports:** Schedule the Python script to run daily and automatically email updated charts and statistics to the marketing team.